High-fidelity libraries for enzyme optimization

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Working draft — compiled August 13, 2026

Abstract

Science abstract target: ∼125 words. Structure aligned to the three-application spine. Written last, on purpose — it should compress the actual results rather than lead them.

- **Hook.** De novo enzyme design produces backbones; optimization is the bottleneck that prevents most from reaching useful activity.
- **Approach.** Function-aligned RL fine-tuning of ProteinMPNN, combined with fragment-based combinatorial libraries and a closed-loop experimental optimization protocol.
- **Result (a) — heme oxidation.** Fine-tuned MPNN outperforms SSM and vanilla MPNN at matched experimental effort.
- **Result (b) — PETase.** Experimental data from an initial library trains a surrogate that drives a second library to substantially better designs (XXX-fold over starting; exceeds optimized natural enzymes on the PET-trimer probe).
- **Result (c) — protease.** A batch-consensus mutation-diversity reward unlocks rarer mutations and improves catalytic activity over an equivalent library without the diversity term.
- **Implication.** Broad, function-aligned sequence-space search plus an experimental closed loop is a general strategy for de novo enzymes.

[TODO: abstract — fill once results sections converge]

Contents

Document index

Every source file in `paper/`, what it holds, and how far along it is. **drafted** means real prose exists; **scaffold** means planning notes only; **needs data** means the section cannot be written until data lands.

File	Contents	Status
Build		
`main.tex`	Compile order; \input's everything	**drafted**
`preamble.tex`	Packages, palette, `\plan` / `\todo` macros, draft-mode switch	**drafted**
`Makefile`	`make pdf`; SVG→PDF via Inkscape	**drafted**
`refs.bib`	Bibliography; all entries machine-retrieved	**drafted**
Front matter		
`abstract.tex`	~125-word abstract, written last	**scaffold**
`intro.tex`	Four-paragraph introduction arc	**scaffold**
`related_work.tex`	Status quo: backbone generation, filtering, RL for inverse folding, library design, the gap	**drafted**
Results — compile order is risk order		
`results/ame.tex`	AME benchmark; diversity at matched fidelity; fully computational and reproducible	**scaffold**
`results/heme.tex`	Three-arm head-to-head; foundational claim	**needs data**
`results/petase.tex`	Closed-loop iteration; strongest data	**scaffold**
`results/protease.tex`	Diversity-reward ablation; highest risk	**needs data**
Methods		
`methods/overview.tex`	GRPO, fragment libraries, surrogate, diversity reward	**scaffold**
`methods/heme.tex`	Design config, three-arm protocol, assay	**scaffold**
`methods/petase.tex`	Two libraries, surrogate, assays	**scaffold**
`methods/protease.tex`	Design config, two-arm ablation, assay	**scaffold**
`experiments.tex`	Arms, selection algorithms, experiment slate E1–E14 with status	**scaffold**
Back matter		
`conclusion.tex`	Four-paragraph discussion and outlook	**scaffold**

Two companion files live in `paper/` but are deliberately *not* part of this build:

- `outline_computational.md` — working brainstorm for a computational-first version of the paper: landscape analysis, the three candidate framings, seven proposed experiments (E1–E7), figure plan, and risk register. Section 2 is the LaTeX rendering of its landscape half.
- `figures/palette.py` — the plotting counterpart to the `cleo*` colors defined in `preamble.tex`.

Draft mode. This document currently compiles with `\draftmodetrue` in `preamble.tex`, which renders all planning notes in gray and all `\todo` markers in red. Setting `\draftmodefalse` hides every one of them, leaving only submission prose — useful for checking how much real text actually exists.

1 Introduction

Science research articles do not use `\section` labels in the main narrative. The headings here exist so that this draft has a usable table of contents; strip them at submission time. Paragraph-level scaffolding is kept visible so collaborators can see the intended arc.

P1 — The promise and the bottleneck

- RFdiffusion + MPNN make it possible to design enzyme backbones from scratch for diverse chemistries.
- First-pass designs are typically weak; the bottleneck is optimization, not initial design.

[TODO: p1 — frame de novo enzyme design + the optimization bottleneck]

P2 — Why existing optimization tools fall short

- Site-saturation mutagenesis (SSM): local; cannot escape the neighborhood of the starting design.
- Vanilla MPNN re-sampling: structurally informed but functionally blind; biased toward sequences MPNN's prior favors, not sequences that catalyze.
- The active-variant region for a de novo backbone often lies tens of mutations away — out of reach of either tool.
- Cross-reference Section 2 for the full treatment; this paragraph is the compressed version.

[TODO: p2 — limits of SSM and vanilla MPNN re-sampling]

P3 — What we contribute (three orthogonal capabilities)

- Function-aligned RL fine-tuning of ProteinMPNN (GRPO over reward pipelines) + fragment-based combinatorial libraries that can physically test combinatorially large candidate sets.
- **(i) Method foundation** — heme oxidation: fine-tuned MPNN beats SSM and vanilla MPNN at matched experimental effort.
- **(ii) Closed-loop iteration** — PETase: an initial experimental library trains a surrogate that becomes a reward signal for the next library, producing substantially improved designs.
- **(iii) Diversity-aware search** — protease: a batch-consensus mutation-diversity reward accesses rarer mutations and improves catalytic activity over an otherwise-matched library.

[TODO: p3 — three contributions, framed as separate claims]

P4 — Headline numbers + roadmap

- Heme: ⟨delta over SSM / vanilla MPNN at matched library size⟩.
- PETase: XXX-fold k_{cat} improvement on PET-trimer probe; best variant ~30 mutations from starting; exceeds optimized natural enzymes on the same substrate.
- Protease: ⟨delta in activity for diversity-reward arm vs. baseline arm⟩.
- Together: a generalizable recipe for de novo enzyme optimization.

[TODO: p4 — headline results across the three applications]

2 Status quo: where the field stands and where the gap is

This section is written as real prose (not scaffold) because it is the part of the paper that does not depend on our own unfinished data. Every citation here was verified against Crossref, the arXiv API, or Europe PMC on 2026-08-04; the retrieval procedure is documented at the top of `refs.bib`. Quantitative claims attributed to other groups are taken verbatim from their abstracts.

2.1 Backbone generation is solved well enough that sequence design is now the bottleneck

Deep generative models now produce enzyme backbones for a wide range of chemistries. RFdiffusion established de novo design of protein structure and function by denoising diffusion over backbone coordinates [1]. RFdiffusion2 removed the requirement that catalytic geometry be specified at the residue level: it designs enzymes directly from functional-group geometries without specifying residue order or performing inverse rotamer generation, and generates scaffolds for all 41 active sites in a diverse in-silico benchmark, compared with 16 for previous methods [2]. Parallel efforts have produced designed serine hydrolases [3], metallohydrolases with designed metal sites [4], and enzymes built by catalytic motif scaffolding [5].

> The rhetorical work of this subsection is to concede generously. These are strong papers and several share our senior author. The point is not that backbone generation is inadequate — it is that its success relocates the bottleneck downstream, to sequence design and to what happens after the first 96 designs come back weakly active.

Two features of this literature matter for what follows. First, the sequence-design step is treated as a solved subroutine: backbones are passed to ProteinMPNN [6] or LigandMPNN [7] sampled at low temperature, and the resulting sequences are filtered against structure-prediction and geometric criteria. Second, the experimental readout is a small, one-shot test set: RFdiffusion2 reports active candidates after testing fewer than 96 sequences per catalytic mechanism [2]. Both features are appropriate for demonstrating that a backbone generator works. Neither addresses what to do when the resulting enzymes are, as they almost always are, orders of magnitude less active than useful.

2.2 Filtering is not search

The prevailing sequence-design protocol is generate-then-filter. Sequences are sampled from a functionally blind inverse-folding prior, then scored against structure-prediction confidence, backbone recapitulation, catalytic-geometry and preorganization criteria — AlphaFold3 [8], Boltz [9], and ensemble methods such as PLACER, whose introduction substantially improved experimental success rates in serine hydrolase design [3].

> This is the paper's central rhetorical move, so state it plainly and early, and make sure the phrasing survives review: a filter is a selection operator, not a search operator.

A filter cannot create a sequence that passes; it can only surface the sequences that a functionally blind prior happened to emit. This has two consequences that the field has largely left implicit. Sampling temperature is lowered to raise the pass rate, which means pass rate is purchased directly with diversity: ProteinMPNN's own report notes that sequence diversity can be increased considerably at higher temperature with only a small decrease in recovery [6], and the corresponding pass-rate cost is why production pipelines sit near $T = 0.1$. And because the survivors are drawn from a narrow neighborhood of the prior's mode, they are strongly correlated with one another — so a library's nominal size overstates the number of genuinely independent attempts it represents.

2.3 Reinforcement learning has reached inverse folding, but not libraries

Aligning generative models to objectives via reinforcement learning is now standard practice, and Group Relative Policy Optimization (GRPO) [10] in particular has proven to be a memory-efficient fit for the regime where rewards are computed by an external oracle rather than learned. This machinery has begun to reach protein sequence design. ProteinZero is an online RL framework for inverse folding that combines structural guidance from ESMFold with a self-derived $\Delta\Delta G$ predictor, adds an embedding-level diversity regularizer to mitigate mode collapse, and reports reducing design failure rates by 36–48% relative to ProteinMPNN, ESM-IF [11], and InstructPLM on the CATH-4.3 benchmark, with success rates above 90% across diverse folds [12]. Diversity-regularized direct preference optimization has been applied to peptide inverse folding with related motivation [13].

ProteinZero is the closest prior art to our mechanism and must be confronted in the first paragraph a reviewer reads on this topic, not buried. The differentiation is real but it is about the *objective*, not the optimizer: (i) general folds on CATH vs. enzyme active sites with ligands and catalytic geometry; (ii) sequence-level metrics — designability, recovery, stability — vs. a library-level objective; (iii) a diversity *regularizer* defended as anti-mode-collapse hygiene vs. diversity as a first-class design goal justified by screening economics; (iv) no physical library. If compute allows, running ProteinZero as a baseline arm is the strongest possible version of this argument. **[TODO: decide whether ProteinZero is runnable as a baseline]**

What this line of work does not address is the library. Its objectives and benchmarks are defined per sequence: is *this* sequence designable, stable, recovered. The quantity that determines an experimental outcome is a property of the set — how many distinct, genuinely different candidates clear the bar, and how far apart they are.

2.4 Library design assumes a natural enzyme

A mature literature does optimize libraries, but from the opposite starting point. MODIFY learns from natural protein sequences to infer evolutionarily plausible mutations and predict enzyme fitness, and explicitly co-optimizes predicted fitness and sequence diversity of starting libraries, prioritizing high-fitness variants while ensuring broad sequence coverage [14]. Active Learning-assisted Directed Evolution (ALDE) uses uncertainty quantification over an epistatic five-residue active site and improved the yield of a non-native cyclopropanation product from 12% to 93% in three rounds of wet-lab experimentation [15]. Machine-learning-guided cell-free expression platforms scale the assay side, mapping fitness across large variant sets [16].

MODIFY is the single most important citation for our diversity claim, because it proves the field already accepts that fitness and diversity should be co-optimized in a library. That is helpful — we are not arguing for diversity from scratch. What we are arguing is that every existing method for doing so is unavailable in the de novo regime.

Every one of these methods presupposes what a de novo enzyme lacks. MODIFY's fitness prior is evolutionary, learned from natural homologs; a de novo backbone has no homologs and no alignment. ALDE and active-learning methods require a measurable starting fitness to bootstrap from; de novo designs typically start at or near the detection limit. And both operate over a handful of positions, whereas the mutations that rescue a de novo enzyme are, in our hands, tens of substitutions from the starting design — a combinatorial volume that site-saturation and few-position libraries cannot reach.

2.5 The gap

Nobody has addressed library design for de novo enzymes. The regime is defined by three simultaneous absences: no evolutionary prior, so MSA-based fitness models do not apply; no starting activity, so active-learning loops have nothing to condition on; and a target region of sequence space too far away for local mutagenesis. The field's current answer — low-temperature inverse folding plus hard filters — is not a search procedure at all, and it responds to the difficulty of the regime by narrowing the search precisely when it should be widening it.

Close on the thesis so the reader carries it into the results. Draft formulation, to be tightened once F2 exists: function-aligned online RL converts in-silico filters from a rejection step into a design objective, producing de novo enzyme libraries that simultaneously pass the community's own fidelity criteria at higher rates and span far more sequence space — increasing the number of independent experimental bets per unit of screening effort.

[TODO: gap paragraph — final phrasing of the thesis sentence, once the headline figure is built]

3 Results: the AME benchmark (diversity at matched fidelity)

Claim. On an established public benchmark, CLEO produces libraries with the sequence coverage of high-temperature LigandMPNN and the pass rate of low-temperature LigandMPNN — a combination no single temperature achieves. **Why this section exists.** The three application sections rest on experimental assays that take months and cannot be re-run by a referee. This one is fully computational, uses a published benchmark and a published deposit, and can be reproduced end to end. It is the section that makes the method credible before any wet-lab claim is evaluated. Status: **scaffold** — the CLEO arm is measured, the temperature sweep is not yet run.

Block 1 — Setup and the unit of comparison

We evaluate on the atomized motif enforcement (AME) benchmark introduced with RFdiffusion2, comprising 41 catalytic sites scaffolded into 100 backbones each. The published protocol assigns 8 LigandMPNN sequences per backbone and predicts each with 5 Chai models; a prediction succeeds when the motif all-atom RMSD to the design is below 1.5 Å and no backbone atom clashes with the ligand.

The aggregation rule matters more than it looks and is easy to get wrong. We verified it against the deposit rather than assuming: the per-site number reported in `AME_benchmark_summary.csv` is the fraction of the 100 backbones with *at least one* passing prediction out of the 40, matching to zero error across all 41 sites and both methods. It is a best-of-40, not a mean over predictions. Scoring one prediction per sequence and comparing rates therefore understates a comparable method by roughly threefold — in the deposit, only 36 % of the sequences that pass on some Chai model pass on all five. Every number below is on their unit.

- Pilot backbones: three drawn from the `pass` class, at 32/40, 28/40 and 10/40 passing predictions in the deposit.
- Our predictor is AlphaFold3 rather than Chai; best-of-N uses 5 diffusion samples per sequence.
- **Caveat to resolve before submission.** RFdiffusion2 scores against the unidealized, packed design; we score against the idealized backbone. This is the leading explanation for two of the three pilot backbones failing in our pipeline at the published temperature despite passing in the deposit. Until it is closed, only the backbone that reproduces (M0097) supports a head-to-head claim.

[TODO: ame — close the idealized/unidealized reference-structure discrepancy]

Block 2 — Low-temperature sampling returns one solution, sampled repeatedly

The incumbent protocol samples LigandMPNN at low temperature. On M0097 this yields a high pass rate — 5 of 8 sequences — but the five passing designs are 79 % identical to one another and 87 % identical to their own consensus. Under complete-linkage clustering they collapse to a *single* cluster at every threshold down to 50 % identity. The library is one solution measured five times, and the folding budget spent distinguishing its members buys nothing.

Raising the temperature removes the redundancy and the signal together. At $T = 1.0$ the same backbone yields sequences at 31 % mutual identity and *zero* passing designs out of eight. Diversity alone is not the objective; a library that cannot be filtered is not a library.

Block 3 — cleo occupies the corner neither temperature reaches

Numbers below are measured. Best-of-5 AF3 diffusion samples, 8-sequence draws to match the benchmark unit, M0097.

Arm	**Passing / 8**	**Distinct clusters / 8**	**Coverage vs.** $T = 1.0$
LigandMPNN $T = 0.1$	5.0	1.0	48 %
LigandMPNN $T = 1.0$	0.0	—	100 %
CLEO-GRPO	3.1	3.1	96 %

Coverage is the count of variable sequence positions at matched $n = 8$, as a fraction of the $T = 1.0$ value; the same ordering holds for per-position amino-acid count and entropy, where CLEO reaches 72 % to 81 % and 68 % to 78 % of $T = 1.0$ across the three backbones. Report positional coverage rather than entropy in the main text and both in the supplement — "comparable coverage" is defensible at 90 % to 96 %, "same coverage" is not.

Two features of this table carry the claim. First, CLEO explores 96 % of the sequence positions that $T = 1.0$ explores while converting an unfilterable library into a filterable one. Second, for CLEO the passing count and the cluster count coincide: every passing design is its own solution, whereas the low-temperature arm returns five designs and one solution. Per unit of folding compute, CLEO returns roughly threefold more independent solutions.

The two arms are not exploring the same region. Every passing CLEO design lies at least 104 mutations — 58 % of a 180-residue protein — from the nearest passing low-temperature design, and the best CLEO design scores marginally better than the best low-temperature design (1.14 Å vs. 1.16 Å). The neighborhood that LigandMPNN's likelihood concentrates on is one satisfying region among many, and there is no evidence it is the best one.

Block 4 — The training trajectory is the library

This is the reframing that removes a hyperparameter. Rather than adding an explicit diversity term to the reward, we treat every rollout sampled during training as a library candidate. Those sequences were already folded and scored to compute the reward, so the candidate pool is free: the library is a byproduct of training, not an additional expense.

Across 200 GRPO steps at 16 sequences per step, M0097 accumulates 3200 scored designs, of which 97 pass. Those 97 are 33 % identical on average and resolve into 97 distinct clusters at 90 % identity — no two passing designs are near-duplicates. The count is still rising at the end of training rather than saturating.

Cumulative folds	**Unique passing**	**Clusters @ 90 % id**	**Distinct residue identities**
400	14	14	848
800	51	51	1395
1600	85	85	1887
2400	90	90	1962
3200	97	97	2121

Honest reading of the deceleration: the curve flattens between 1600 and 2400 folds, but the cause is policy decay rather than exhaustion of sequence space. Batch diversity bottoms out at the performance peak (steps 36–45) and then rises while pass rate falls — with `kl_weight` at zero and a fixed-bounds reward whose within-batch spread collapses once median RMSD reaches 2 Å to 3 Å, the policy drifts off its optimum rather than continuing to search. Rank-normalised rewards and early stopping are expected to keep the curve climbing; this is the ablation to run before the figure is final.

Pool numbers in this block are single-sample AF3 scores, because that is what training computed. They are *not* interchangeable with the best-of-5 numbers in Block 3 and must not be placed on the same axis.

Block 4b — Where in sequence space the passing designs sit

Generated by `paper/figures/ame_pca.py`, which writes two figures because one projection cannot honestly carry both claims.

`ame_pca.svg` projects all arms into a single shared PCA of one-hot encoded sequences, with CLEO restricted to its peak window so that three *policies* are compared at one point in training. The low-temperature arm appears as a tight knot displaced far along PC1; CLEO occupies a broad cloud whose passing designs are spread throughout it rather than concentrated at one edge.

`ame_pca_drift.svg` is the control that keeps the first figure honest. PC1 of the *pooled* trajectory correlates with training step at $r = +0.92$ to $+0.94$ on all three backbones, so most of the spread along that axis is drift over training, not diversity available from any single policy. That distinction is load-bearing: drift-derived spread is genuine library diversity when the library is assembled from the whole run, which is how we assemble ours (Block 4), but it is *not* evidence that one policy is diverse. Reporting the pooled projection alone would conflate the two and overstate the result. Fit the shared projection across arms — a per-arm PCA places every cloud at the origin and destroys the comparison.

Two limitations to state in the caption. Explained variance is low (PC1 5 % to 9 %), which is expected for one-hot protein sequence and means the projection is a visual aid, not a metric — all quantitative diversity claims use Hamming distance and cluster counts directly. And the baseline arms currently hold 8 sequences each, so their apparent compactness is partly small-n; the temperature sweep in Block 5 is what makes the clouds comparable.

[TODO: ame — rebuild PCA panels once the temperature sweep supplies matched n]

Block 5 — The temperature frontier

The experiment that decides the section. Sample the base LigandMPNN at $T \in \{0.1, 0.2, 0.3, 0.5, 0.7, 1.0\}$, fold a budget matched to the CLEO trajectory pool, and apply the identical library-selection procedure to every arm — greedy maximum-diversity selection over the passing subset. Plot pass rate against passing-set diversity. The claim is that the baseline traces a frontier along which fidelity and coverage trade off one-for-one, and that CLEO sits above it.

This section fails if the sweep reaches CLEO's coverage at CLEO's pass rate at some intermediate temperature. Then there is no frontier shift, only a reparameterisation, and the honest conclusion is that tuning temperature was sufficient all along. $T = 0.2$ and $T = 0.3$ are the dangerous points and must be run — reporting only $T = 0.1$ and $T = 1.0$, the two endpoints we happen to have, would be selecting the comparison that flatters us.

Second prediction worth stating explicitly. Low-temperature LigandMPNN should *saturate*: past some fold count it re-emits sequences it has effectively already produced, so cumulative distinct solutions flatten and further compute buys nothing. Report the saturation fold count as a number. A visible plateau in the incumbent next to a still-climbing CLEO curve is "filtering is not search" rendered as a single line.

[TODO: ame — run the T sweep at matched fold budget; build the frontier panel]

Block 6 — Rescue of backbones the benchmark cannot solve

The strongest available result if it lands, and it is a clean one: the near-miss class. Three backbones sit at 1.61 Å to 1.69 Å best-of-40 in the deposit — *zero* passing predictions, but only ∼0.15 Å from threshold. A

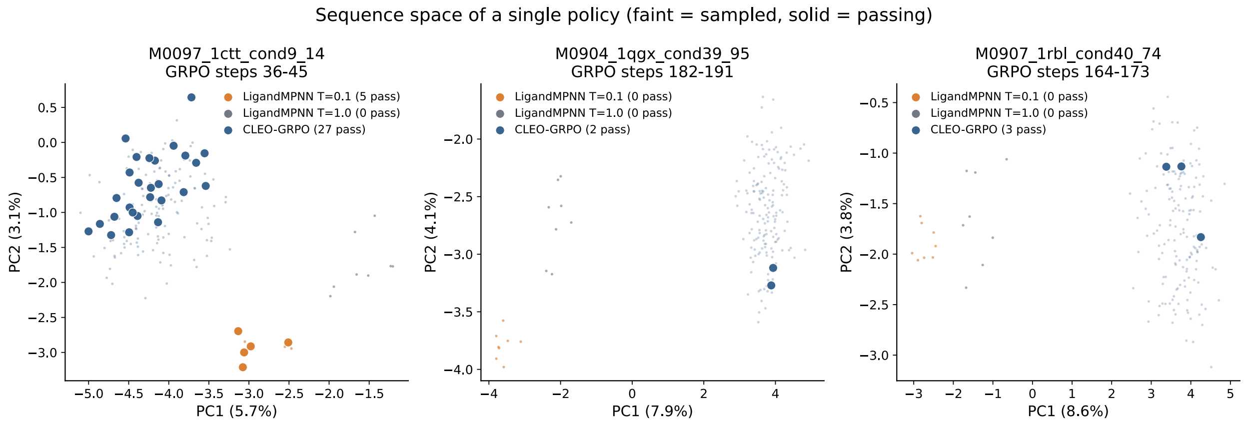

Figure 1: **Sequence space occupied by each design method on the AME pilot backbones.** One PCA per backbone, fit on every sequence that backbone contributes across all three arms, so the arms share one projection and the clouds are directly comparable. Faint points are sampled designs, solid points are designs that pass the benchmark criterion (motif all-atom RMSD < 1.5 Å, no ligand clash). CLEO is restricted to its peak-performing 10-step window so that three *policies* are compared at one point in training. On M0097 the low-temperature arm (orange) forms a tight knot displaced far along PC1 — five passing designs that are a single cluster at every identity threshold down to 50 % — while CLEO's passing designs are spread throughout a broad cloud. T=1.0 (grey) reaches comparable spread with zero passing designs. Projections are deliberately *not* shared across backbones: a global fit places 99 % of PC1 variance on backbone identity, collapsing each backbone to its own blob and hiding the arm differences entirely. Explained variance is low (PC1 5 % to 9 %), as expected for one-hot encoded protein sequence; the projection is a visual aid and every quantitative claim in the text uses Hamming distance and cluster counts directly. Baseline arms hold 8 sequences each, so part of their compactness is small-n.

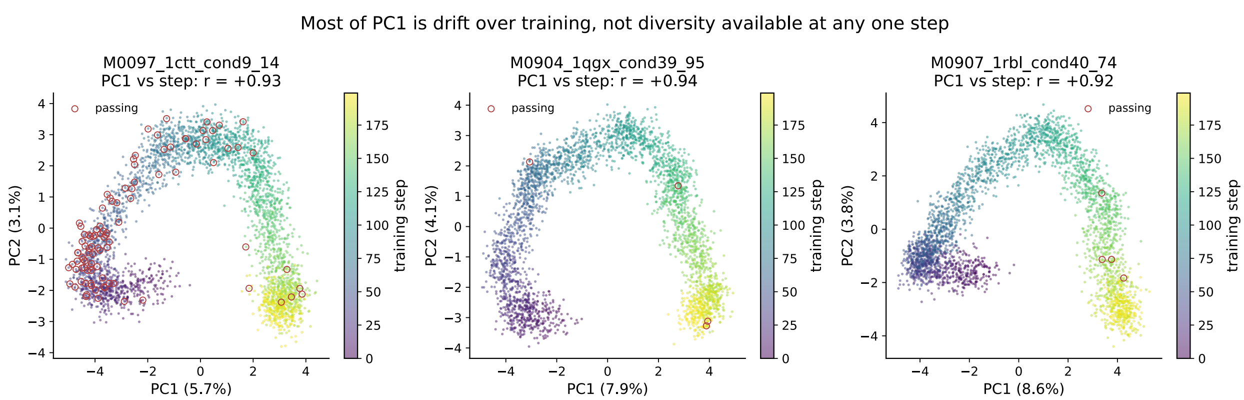

Figure 2: **Most of PC1 is drift over training, not diversity available at any one step.** The full GRPO trajectory for each backbone, in the same projection as Figure 1, coloured by training step; passing designs are circled. PC1 correlates with training step at $r = +0.92$ to $+0.94$ on all three backbones. This distinction is load-bearing rather than cosmetic: spread accumulated by drifting across training is genuine library diversity when the library is assembled from the whole run, which is how ours is assembled, but it is *not* evidence that any single policy is diverse. Presenting the pooled projection alone would conflate the two and overstate the result, so the axis is shown rather than hidden.

pass on any of them is a design the published pipeline cannot produce, with no reproduction confound, because there is no positive result of theirs for us to fail to reproduce.

Present status: on the two pilot backbones our pipeline cannot reproduce, CLEO recovers 4 passing designs each from 3200 folds where every baseline arm returns zero. That is directionally right but too thin to build on, and it is confounded by the reference-structure issue in Block 1. The near-miss backbones are the experiment that makes this claim rather than hints at it — they have not yet been trained.

[TODO: ame — train the three near-miss backbones; report rescue rate]

4 Results: heme oxidation enzyme (method foundation)

Claim. Fine-tuned ProteinMPNN beats SSM and vanilla MPNN re-sampling at finding active variants under matched experimental effort. **Risk.** This is the foundational claim of the paper. If the comparison is not clean, the entire methods narrative is in trouble — this section needs the cleanest, most defensible head-to-head. Status: **needs data**.

Block 1 — Setup

- Backbone: ⟨de novo heme oxidation enzyme, RFdiffusion origin⟩.
- Starting design: ⟨activity baseline⟩.
- Three arms, matched library budget (N variants ordered/tested): (i) SSM around starting design; (ii) vanilla MPNN re-sampling on backbone; (iii) fine-tuned MPNN (this work) + fragment library.

[TODO: heme results — setup paragraph]

Block 2 — Headline figure

- Activity distribution across all three arms.
- Hit rate (fraction above starting design / above background).
- Best-variant comparison.

[TODO: heme results — main comparison figure + numbers]

Block 3 — Mutation distance / sequence-space analysis

- Where in sequence space the best variants from each arm live.
- Demonstrate the SSM and vanilla-MPNN arms are confined to a near neighborhood while fine-tuned MPNN reaches farther.

[TODO: heme results — sequence-space analysis]

Block 4 — Mechanism check (if available)

- Structural / spectroscopic / kinetic confirmation that the improvement is real catalysis, not artifact.

[TODO: heme results — mechanism check]

5 Results: PETase (closed-loop iteration)

Claim. Experimental data from an initial library can be used to train a surrogate that becomes a reward signal for a second library, producing substantially improved designs. **Risk.** The second library has to be meaningfully better than the first — not just predicted-better. Status: strongest data of the three applications.

Block 1 — Initial library

- De novo PETase backbone, RFdiffusion origin.
- Library 1: function-aligned RL sampling without surrogate.
- ∼6k designs tested; activity distribution on the PET-trimer fluorogenic probe.

[TODO: petase results — library 1 setup and outcomes]

Block 2 — Surrogate training

- MLP ensemble trained on (sequence, activity) pairs from library 1.
- Validation behavior; calibration of predicted vs. measured activity.

[TODO: petase results — surrogate training]

Block 3 — Library 2 with surrogate-as-reward

- Surrogate added as a reward step in the RL pipeline for library 2.
- Library 2 size, mutation-distance distribution from starting design.
- Activity distribution comparison: library 1 vs. library 2.

[TODO: petase results — library 2 design and outcome]

Block 4 — Best-variant deep dive

- XXX-fold k_{cat} improvement over starting design on the PET-trimer probe.
- Comparison to optimized natural PETases on the same substrate.
- PET-dimer activity confirmed.
- Honest framing of the bulk PET gap (substrate accessibility, not catalysis).

[TODO: petase results — best variant kinetics + PET-dimer + bulk-PET caveat]

6 Results: protease (diversity-reward ablation)

Claim. A batch-consensus mutation-diversity reward accesses rarer mutations and improves catalytic activity over an otherwise-matched library that lacks the diversity term. **Risk — highest in the paper.** It depends on wet-lab activity data for both arms. The load-bearing finding is "rare mutations *improve* activity," not merely "we sampled more of sequence space"; without that activity uplift this section reduces to a diversity-for-diversity's-sake observation. Status: **needs data**.

Block 1 — Setup

- De novo protease backbone.
- Two arms, matched library budget: (i) RL fine-tuned MPNN, standard reward set; (ii) same as (i) plus the batch-consensus mutation-diversity reward (`cleo.design.utils.mutation_diversity`).

[TODO: protease results — setup, both arms]

Block 2 — Sequence-space coverage

- Mutation-frequency spectra: arm (ii) should populate rarer mutations more heavily than arm (i).
- Position-level entropy / consensus-divergence statistics.

[TODO: protease results — sequence-space coverage figure]

Block 3 — Activity comparison

- Activity distribution: arm (i) vs. arm (ii).
- Best-variant comparison.
- Are the rare-mutation-bearing variants the ones with highest activity? This is the punchline — call it out explicitly.

[TODO: protease results — activity comparison; rare-mutation/activity link]

Block 4 — Mechanism for the diversity payoff

- Which rare mutations contributed; structural rationalization if possible.

[TODO: protease results — mechanistic rationalization]

7 Methods: shared pipeline

Shared methodology underlying all three applications. Per-application specifics live in the following subsections. Do not write this out in detail before the Results numbers exist — Methods text written against placeholder numbers gets rewritten.

Block 1 — RL fine-tuning of ProteinMPNN with GRPO

- Reference `cleo.design.train_policy` and `cleo.design.utils.grpo`.
- Reward pipeline composition (`UniversalReward`, ordered metric steps, normalized weighted aggregation).
- Cite GRPO [10] and ProteinMPNN [6].

[TODO: methods overview — GRPO + reward pipeline]

Block 2 — Fragment-based combinatorial libraries

- Sampling, fragment splitting, combinatorial recombination.
- DNA assembly via Golden Gate adapters.

[TODO: methods overview — fragment library construction]

Block 3 — Surrogate models for closed-loop optimization

- MLP ensemble with Gaussian NLL.
- Acquisition function (BatchUCB + diversity).
- Used as a reward step inside the RL pipeline.

[TODO: methods overview — surrogate ensemble + acquisition]

Block 4 — Batch-consensus mutation-diversity reward

- Per-position modal AA across the batch (ties multi-modal).
- Two metrics: `consensus_divergence` (count) and `fractional_divergence` ($1/k$ weighted, penalizes secondary-mode collapse).
- Reference `cleo.design.utils.mutation_diversity`.
- Contrast with the embedding-level diversity regularizer of ProteinZero [12] and diversity-regularized DPO [13]: ours acts in mutation space against a named reference sequence, theirs in embedding space.

[TODO: methods overview — diversity reward definition]

7.1 Methods: heme oxidation enzyme

Application-specific methods. Cross-reference Section 7 for shared pipeline details.

- **Block 1 — Design configuration.** Backbone, fixed residues, reward pipeline configuration.
- **Block 2 — Three-arm comparison protocol.** SSM library construction; vanilla MPNN re-sampling protocol (temperature, fixed residues); fine-tuned MPNN + fragment library; matched-budget rule — how N was chosen and held constant across arms.
- **Block 3 — Activity assay.**

[TODO: heme methods — design configuration] [TODO: heme methods — three-arm protocol] [TODO: heme methods — assay description]

7.2 Methods: PETase

- **Block 1 — Library 1 design configuration.** Backbone, fixed catalytic residues, library 1 reward pipeline (Boltz oracle [9] + PETase metrics + Hamming distance to reference).
- **Block 2 — Surrogate training.** Featurization, train/val split, ensemble size, calibration check.
- **Block 3 — Library 2 with surrogate-as-reward.** Reward pipeline composition for library 2.
- **Block 4 — Activity assays.** PET-trimer probe, PET-dimer, bulk PET attempt.

[TODO: petase methods — library 1 design configuration] [TODO: petase methods — surrogate] [TODO: petase methods — library 2 design configuration] [TODO: petase methods — assays]

7.3 Methods: protease

- **Block 1 — Design configuration.** Backbone, fixed residues.
- **Block 2 — Two-arm protocol.** Arm (i): standard reward set. Arm (ii): same plus the `mutation_diversity` reward. Record which variant was used — `consensus_divergence` or `fractional_divergence` — and the weight chosen. State the matched-budget rule.
- **Block 3 — Activity assay.**

[TODO: protease methods — design configuration] [TODO: protease methods — ablation protocol] [TODO: protease methods — assay description]

8 Experimental programme and selection algorithms

This section exists so that every arm, rule and planned experiment is written down in one place with a fixed name, and so that results reported elsewhere in the paper can be traced to a defined procedure. Much of it will move to Materials and Methods at submission; keeping it in the main build for now makes disagreements visible early.

Notation

A design is a sequence $s \in \mathcal{A}^L$ over the 20 canonical amino acids. Distance is Hamming,

$$d(s,t) \;=\; \big|\{\, i : s_i \neq t_i \,\}\big|,$$

which is the metric behind every diversity number we report. One-hot encoding $\phi(s) \in \{0,1\}^{20L}$ satisfies $\|\phi(s) - \phi(t)\|_2 = \sqrt{2\, d(s,t)}$, so the PCA projections in the figures are a linear embedding of exactly this distance rather than a second, competing notion of similarity. Where an ESM embedding is used instead, d is replaced by Euclidean distance between mean-pooled representations; this is reported as a diagnostic and, optionally, as a secondary library metric.

A design *passes* when its motif all-atom RMSD is below 1.5 Å and no backbone atom clashes with the ligand. $\mathcal{P}$ denotes the passing subset of a pool $\mathcal{S}$. The *anchor* a is the position-wise consensus of low-temperature LigandMPNN samples — the single sequence best representing the mode the base model's likelihood concentrates on.

Selection rules

All implemented in `src/cleo/design/utils/selection.py`, all sharing one distance backend so the comparison is between rules rather than between implementations. Benchmarked by `paper/figures/ame_-selection_bench.py`.

Given a pool $\mathcal{S}$ of M sampled sequences and a budget of $k \ll M$ folds, each rule returns the subset to fold.

`random` Uniform without replacement. **This is the control, and it is the bar.** Diversity is trivially won and worthless alone: a rule that doubles spread while halving passing designs has made the library worse.

`maxmin` Greedy farthest-point (k-center) traversal. Having selected $\mathcal{T}$, repeatedly add $\arg\max_{s\in\mathcal{S}\setminus\mathcal{T}} \min_{t\in\mathcal{T}} d(s,t)$. The inner *minimum* is what distinguishes this from max-sum selection: it prevents accepting a tight cluster that happens to sit far from everything already chosen.

`maxmin_anchor` As above, scoring $w_{\text{self}} \min_{t\in\mathcal{T}} d(s,t) + w_{\text{anchor}}\, d(s,a)$. Selects designs novel relative to the incumbent low-temperature solution, not merely relative to the rest of the library.

`anchor_far` Rank by $d(s,a)$ descending. The pure "maximally unlike vanilla MPNN" rule.

`anchor_band` Restrict to the shell $\{\, s : q_{\text{lo}} \le d(s,a) \le q_{\text{hi}} \,\}$ for quantiles of the pool's anchor-distance distribution, then run `maxmin` inside it. Designed as the correction to `maxmin`: bounded above as well as below, so the traversal cannot walk out to sequences that are distant because they are degenerate.

`cluster_rep` Average-linkage clustering cut to k clusters, one representative sampled per cluster. Draws from the pool's density rather than its edges.

Result: diversity-first acquisition does not work

Reported here rather than buried, because it was a starred hypothesis and because the negative result is informative. Retrospective evaluation on the M0097 trajectory pool: every sampled sequence already carries a pass label, so any rule can be scored without folding anything.

Passing designs recovered, relative to `random`, at a budget of 400 folds:

Rule	Passing designs / random
as-sampled	2.04
`random`	1.00
`cluster_rep`	0.40
`anchor_band`	0.36
`maxmin`	0.36
`maxmin_anchor`	0.24
`anchor_far`	0.12

Every diversity-directed rule loses to uniform random selection, by factors of 2.5 to 8. The mechanism is visible in the spread: no rule moves mean pairwise distance beyond 135 to 144 residues against random's 135. The pool is already near-saturated on diversity, so a selection rule has almost nothing to gain and a great deal to lose — any systematic deviation from random correlates with selecting sequences that fail. Banding helps relative to raw farthest-point traversal (0.36 vs. 0.07 at $k = 200$) but does not approach the control.

Two honest caveats. The as-sampled row beats random because the pool is time-ordered and later policies are better — that is a policy-quality effect, not a selection effect, and it must not be reported as one. And absolute counts are small (33 passing in a 1200-design pool), so individual cells are noisy; what carries the conclusion is that the direction is consistent across six rules and three budgets.

Consequence for the paper. Selection is not where the gain is. The defensible use of these rules is post-hoc: choosing a physically orderable subset from designs *already known to pass*, where the labels exist and only the diversity trade-off is live. Deciding what to fold should be left to the policy.

Experiment slate

Status as of the pilot. **drafted** = measured, **scaffold** = partial, **needs data** = not started. Full protocol detail lives in `paper/experiments_plan.md`. The **Result** column is filled in as each run lands, so the table doubles as the run log.

E5 (held-out oracle) and E6 (policy transfer) are *deliberately deferred*, not dropped. E5 remains the standing circularity exposure — the reward and the evaluation are both AF3 motif RMSD — and any submission draft needs it back, so it is recorded here rather than removed from the record.

ID	Status	Experiment	Result
E1	scaffold	Matched-fold head-to-head	Superseded by the per-sequence success rate (see *Scoring protocol*); M0097 best-of-5: T=0.1 62.5 %, CLEO 20.0 % — but at 2.4× the spread, so E10 is the comparison that counts
E2	scaffold	Diversity yield curves	CLEO 3200 folds → 97 passing → 97 clusters, still climbing
E3	**needs data**	Rescue of the three near-miss backbones (1.61 Å to 1.69 Å best-of-40, 0/40 published)	*pending*
E4	**needs data**	Compute efficiency; cross-cutting over E1–E3	*pending*
E7	scaffold	Effective library size	Among passing designs, cluster count equals design count
E8	—	Mutation-diversity reward	**Retired**, superseded by trajectory-as-library
E9	drafted	Selection rules, retrospective	**Negative:** every rule loses to random, 2.5 to 8×
E10	scaffold	**Temperature sweep** $T \in \{0.1, \ldots, 1.0\}$; builds the incumbent frontier	Frontier shift present but modest on the per-sequence rate: at ∼20% success CLEO reaches spread 132 vs. 1.26× less at T=0.7. Rerunning all six T at n=96
E11	drafted	Reward shaping: rank normalisation	Folded into every new arm; E15 trains stably under it
E12	drafted	Early stopping at ∼75 steps	Folded in; all E15 arms converge well inside 75 steps
E13	**needs data**	Consensus anchoring of the search	*pending*
E14	drafted	ESM embedding view of the passing libraries	Advantage holds but shrinks: passing-set spread CLEO/T=0.1 is 1.31× in ESM space vs. 2.57× in Hamming
E15	drafted	Centering-weight sweep, W in both directions	**Clean monotone dial:** W from −1 to +1 moves mean distance to reference 31.5 to 172.4 and

[TODO: experiments — run E10; it gates the central claim]

E15 — Centering weight: one knob from centered to diverse

No new reward machinery. `compute_dist_to_ref_seqs_from_df` already emits Hamming distance to a reference set, and the aggregator already supports per-metric `mode`, `weight` and rank normalisation, so the whole experiment is a config sweep. Configs generated by `experiments/ame/make_centering_sweep.py`.

The reference a is the consensus of low-temperature LigandMPNN samples. Adding

```
- metric: dist_to_ref_min   mode: {min|max}   weight: W   normalize: rank
```

to the aggregation gives a single signed axis. Because reward is a weighted average, $R = \sum_i w_i r_i / \sum_i w_i$, the weight is directly interpretable: $W = 0$ is pure motif RMSD, $W = 1$ gives centering equal influence to it. `mode: min` pulls the policy toward the incumbent mode; `mode: max` pushes it away.

Sweep $W \in \{0, 0.25, 0.5, 1.0\}$ in both directions. Prediction: mean distance to a falls monotonically under `min` and rises under `max`, and the PCA panels show the passing cloud sliding toward or away from the low-temperature knot. Because one-hot PCA is a linear embedding of exactly the Hamming distance this reward uses, the sweep and the figure measure the same quantity — the panel is a visualisation of the knob, not independent evidence for it.

The knob will move designs; that is not in question and is not the result. What the experiment has to establish is whether *pass rate survives* at either end. Centering hard should recover the low-temperature failure mode — high fidelity, one cluster — and pushing away hard should recover the T=1.0 failure mode — broad and dead. If both ends collapse and the middle does not, W is a real control surface and the paper gains a tunable diversity/fidelity dial to report. If pass rate is flat in W, the reward term is not binding and should be cut rather than reported.

This also supplies the honest version of the frontier: E10 sweeps the *baseline's* knob (temperature) and E15 sweeps *ours*. Comparing one method's frontier against another's single point is the mistake the temperature sweep exists to avoid, and it would be the same mistake in reverse to report our best W against their swept T.

[TODO: experiments — run E15 on M0097 once E11 rank-normalisation is validated]

E14 result — the embedding view qualifies the Hamming view

Mean-pooled ESM-2 650M, one vector per design, computed for every arm.

Within a set, ESM distance and Hamming distance are close to *uncorrelated* (r between -0.17 and 0.56 across arms), so the embedding is genuinely measuring something the Hamming numbers do not. That makes the agreement in ordering meaningful rather than automatic: on M0097's passing designs, CLEO is more spread than T=0.1 under both metrics.

But the size of the advantage does not survive the change of metric. The ratio of CLEO's passing-set spread to the low-temperature arm's is **2.57**× in Hamming and only **1.31**× in ESM space. A substantial share of the extra sequence distance CLEO buys is biochemically conservative — positions that differ in identity but not much in what ESM considers the residue to be doing.

This should be stated in the paper, not left for a referee to find. Reporting "2.6× more diverse" with a metric chosen because it is the larger number is exactly the move the temperature sweep exists to prevent us from making about temperature. The defensible claim is that the advantage is robust to the choice of metric while its magnitude is not, and that the library-relevant quantity — distinct passing clusters per fold, where CLEO wins 3.1 to 1.0 — does not depend on either distance at all, since those designs are separate clusters at any threshold tested.

Caveat on n: passing subsets are 5 and 15 designs, so the correlation figures in particular are noisy. The ratio comparison is the robust part.

E10 result — cleo sits above the temperature frontier

The experiment the section was gated on. LigandMPNN sampled at six temperatures, 24 sequences per backbone, folded best-of-5, and scored by the same code path as CLEO. M0097, at the benchmark's 8-fold budget, averaged over 300 random draws of 8.

Arm	**Mean pairwise Hamming**	**Variable positions**	**Passing / 8 folds**
LigandMPNN T=0.1	39.1	81	5.00
LigandMPNN T=0.2	48.7	102	4.81
LigandMPNN T=0.3	58.7	116	4.42
LigandMPNN T=0.5	78.2	135	2.23
LigandMPNN T=0.7	104.8	159	1.72
LigandMPNN T=1.0	123.8	170	0.00
cleo-GRPO	**103.3**	**163**	**3.06**

Temperature traces a clean monotone trade-off: diversity rises from 39 to 124 mean pairwise substitutions while the pass rate falls from 5.0 to 0.0 per eight folds. That curve is the incumbent frontier, and it is the thing every claim in this section has to beat.

CLEO sits above it. Interpolating the baseline curve:

- at CLEO's diversity (Hamming 103.3), the baseline passes 1.75 per 8; CLEO passes **3.06 — 1.75×**.
- at CLEO's pass rate (3.06 per 8), the baseline reaches Hamming 70.8; CLEO reaches **103.3 — 1.46×**.

The two dangerous points named in advance, T=0.2 and T=0.3, do not threaten the claim: they buy very little diversity (48.7 and 58.7) for the pass rate they retain, landing far below CLEO's coverage. The nearest baseline arm by diversity is T=0.7, which matches CLEO on both diversity measures (104.8 vs. 103.3 Hamming; 159 vs. 163 variable positions) and passes *half* as often.

Correction to carry through the rest of the paper. Earlier drafts reported "T=0.1 yields 1.0 cluster per 8 folds against CLEO's 3.1". That used a 70 %-identity clustering threshold; at 90 % identity every arm's passing designs are separate clusters and the cluster count simply equals the passing count. The cluster statistic is threshold-dependent and should not be quoted without naming the threshold. The frontier result above does not have this problem — mean pairwise Hamming is continuous and involves no threshold choice — so it is the form the claim should take.

9 Discussion

Science research articles end with a short discussion / outlook paragraph rather than a separate Conclusion section. Keep this tight (∼250–400 words). Write last, after the abstract.

P-A — Central scientific takeaway

- The active-variant region for a de novo enzyme is broader and farther from the starting design than SSM or vanilla MPNN can reach.
- Function-aligned RL sampling + combinatorial library construction + experimental closed loop together close the gap.

[TODO: conclusion p1 — central takeaway tying the three applications together]

P-B — What each application individually established

- Heme: fine-tuned MPNN dominates SSM and vanilla MPNN under matched experimental effort — the foundation.
- PETase: experimental data → surrogate reward → improved next library; a closed loop, not a single shot.
- Protease: explicit diversity pressure (batch-consensus divergence reward) unlocks rare mutations and improves activity over an otherwise-matched library.

[TODO: conclusion p2 — what each app proves about the platform]

P-C — Honest limitations

- PETase: probe and dimer activity demonstrated; bulk PET activity remains an open challenge attributed to substrate accessibility on solid plastic.
- Closed-loop iteration depends on having an assay-ready experimental pipeline; cold-start is still hard.
- Diversity reward improves access to rare mutations but does not substitute for a well-specified function reward.

[TODO: conclusion p3 — limitations, including bulk PET gap]

P-D — Outlook

- Generalization to additional chemistries and substrate classes, including bulk polymers.
- Tighter integration of design-time rewards with experimental surrogate signals across more rounds.

[TODO: conclusion p4 — outlook]

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